



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

2002
HIGHER SCHOOL CERTIFICATE
JUNE EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time — 5 minutes
- Working time — 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks — 100

- Attempt questions 1–6
- All questions are not of equal value, the mark value is shown beside each part.

Examiner: E.Choy

Note: This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Total marks – 100

Attempt Questions 1–6

All questions are not of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (20 marks) Use a SEPARATE writing booklet.

(a) Find $\int \frac{1}{x \ln x} dx$. **2**

(b) Find $\int_0^{\pi/3} \sin^3 x \cos x dx$. **3**

(c) By completing the square, find $\int \frac{dx}{\sqrt{x^2 + 4x + 8}}$. **3**

(d) Use integration by parts to evaluate $\int_0^{\frac{1}{2}} \cos^{-1} x dx$. **3**

(e) (i) Use partial fractions to show that: **3**
$$\int_0^1 \frac{dx}{(x+2)(2x+1)} = \frac{1}{3} \ln 2.$$

(ii) Hence evaluate $\int_0^{\pi/2} \frac{3}{4+5 \sin x} dx$. **3**

(f) Show that $f(x) = x^8 \sin x$ is an odd function. **3**
Hence evaluate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^8 \sin x dx$.

Question 2 (20 marks) Use a SEPARATE writing booklet.

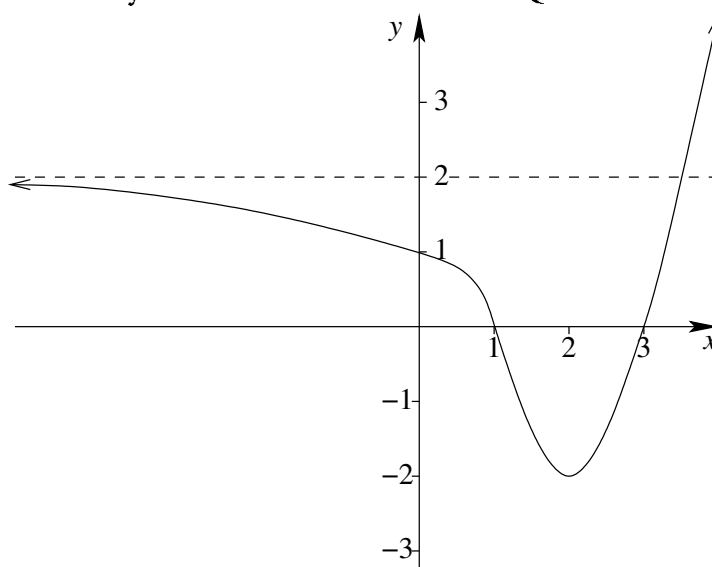
- (a) (i) Simplify i^{2002} . 2
- (ii) Solve $2z^2 + (3+i)z + 2 = 0$. 2
- (b) On separate Argand diagrams, shade the regions:
- (i) $-2 < \text{Im}(z) \leq 5$ 2
- (ii) $|z| < 6$ 2
- (iii) $2 < z + \bar{z} < 10$ 2
- (iv) $\arg(z^2) = \frac{2\pi}{3}$. 2
- (c) In how many ways can 10 women be directed into two groups of 3 and 7 respectively? 2
- (d) If α, β, γ are the roots of the equation $x^3 - 2x^2 + 2x - 2 = 0$, find the value of $\alpha^2 + \beta^2 + \gamma^2$. Explain why only one root of the equation is real. 3
- (e) A certain polynomial, $P(x)$, is an odd polynomial of degree 5. It is given that $P(1) = P(2) = 0$ and $P(3) = 240$. Find $P(x)$. 3

Question 3 (15 marks) Use a SEPARATE writing booklet.

- (a) Solve, graphically or otherwise, $|x^2 - 2x - 3| < 3x - 3$. 4

- (b) On the same set of axes, sketch and label the graphs with equations $y = x(x-3)^2$ and $y^2 = x(x-3)^2$. Clearly indicate turning points and any other critical points. 4

- (c) The diagram below shows the graph of a function, $y = f(x)$. There is an horizontal asymptote $y = 2$ as shown. For your convenience this graph is reproduced on a separate sheet. Using the given graphs as a guide, sketch the required graphs on the separate sheet. Insert the sheet into your examination booklet for Question 3.



- (i) $y = f(x+2)$,
- (ii) $y = |f(x)|$,
- (iii) $|y| = f(x)$,
- (iv) $y = \frac{1}{f(x)}$,
- (v) $y = \ln f(x)$.

Question 4 (15 marks) Use a SEPARATE writing booklet.

(a) Use the substitution $x = \sin^2 \theta$ to evaluate $\int_0^{\frac{1}{2}} \frac{\sqrt{x}}{(1-x)^{3/2}} dx$. 3

(b) (i) If $f(x) = f(a-x)$, prove that 4

$$\int_0^a xf(x) dx = \frac{a}{2} \int_0^a f(x) dx.$$

(ii) Hence or otherwise, prove that $\int_0^\pi g(x) dx = \frac{\pi^2}{4}$, 3

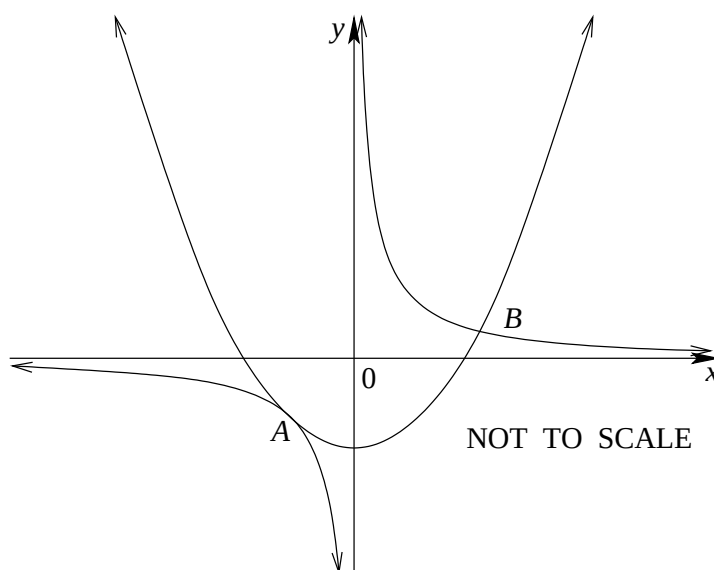
if $g(x) = \frac{x \sin x}{1 + \cos^2 x}$.

(c) (i) Given $I_n = \int_0^1 x^n e^{2x} dx$, where n is a positive integer, use integration by parts to show that $I_n = \frac{1}{2}(e^2 - nI_{n-1})$. 3

(ii) Hence evaluate $\int_0^1 x^4 e^{2x} dx$. 2

Question 5 (15 marks) Use a SEPARATE writing booklet.

(a)



The sketch above shows the graphs of $y = x^2 - b$ and $y = \frac{k}{x}$, where $b > 0$ and $k > 0$. The hyperbola touches the parabola at the point A and cuts it at the point B .

- (i) Show that the x -coordinates of the points of intersection of the hyperbola and the parabola are the roots of the equation $x^3 - bx - k = 0$. 2
 - (ii) Explain why this equation has a double root. 2
 - (iii) Show that $4b^3 = 27k^2$. 3
 - (iv) If $b = 12$, find the coordinates of A and B . 3
- (b) In a hat are 10 red, 10 blue, and 10 yellow cards. Each colour group is numbered from 1 to 10. Four cards are chosen at random from the hat.
- (i) How many groups of four cards can be chosen which contain at least one red card? 2
 - (ii) Find the probability that a group of four cards chosen at random contains at least one of each colour, given that the group contains at least one red card. 3

Question 6 (15 marks) Use a SEPARATE writing booklet.

A particle of unit mass, initially at rest at the origin, is moving along a straight line. The particle is attracted by two objects that are to the right of the origin, one at position A and the other at B . The magnitude of the force due to the object at A is equal to the distance of the particle from A while the magnitude of the force due to B is equal to the square of the distance of the particle from B . Position A is 3 metres from the origin and B is 6 metres from the origin.

- | | | |
|------------------------------|---|----------|
| (i) | Show that the acceleration of the particle for $0 \leq x \leq 3$ and also for $3 \leq x \leq 6$ is given by: | 3 |
| $\ddot{x} = x^2 - 13x + 39.$ | | |
| (ii) | Find an expression for v^2 , the square of the velocity at position x where $0 \leq x \leq 6$. | 2 |
| (iii) | Explain why the particle never comes to rest between the origin and B . | 2 |
| (iv) | Show that the speed of the particle when it first arrives at B is 12 m/s. | 2 |
| (v) | Find an expression for the acceleration of the particle when it is beyond B . | 2 |
| (vi) | Find an expression for the speed of the particle when it is beyond B and explain why the particle comes to rest somewhere between 11 and 12 metres from the origin. (You do not have to find the exact position.) | 4 |

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Name _____

Class _____

GRAPHS FOR QUESTION 3

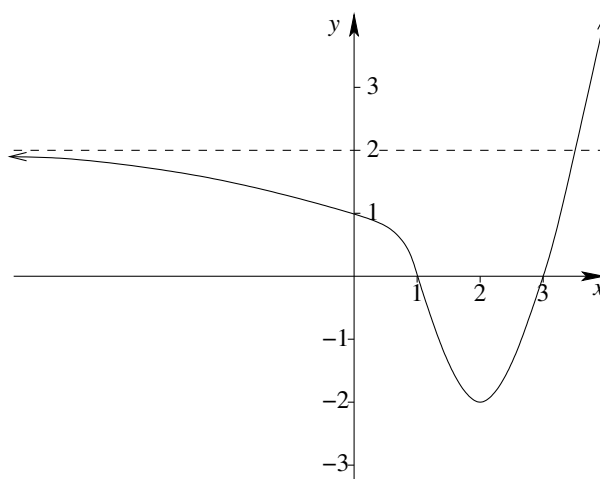
The diagrams on this sheet each show a graph of the function $y = f(x)$, as shown on page 4 of your question booklet. Using the given graphs as a guide, sketch the required graphs.

Insert this sheet into your answer booklet for Question 3.

Marks

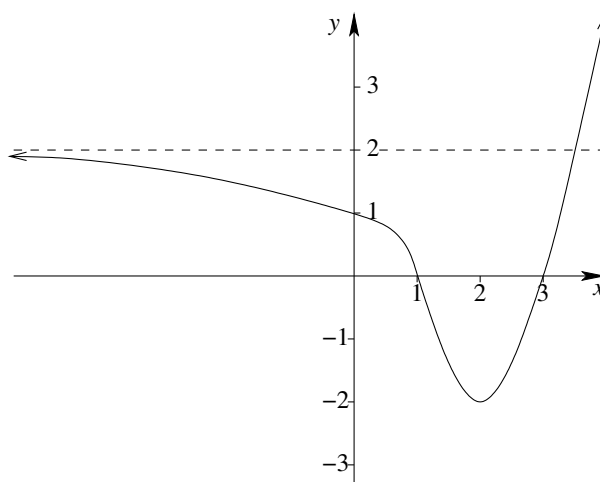
(vii) Sketch $y = f(x+2)$,

2



(viii) Sketch $y = |f(x)|$.

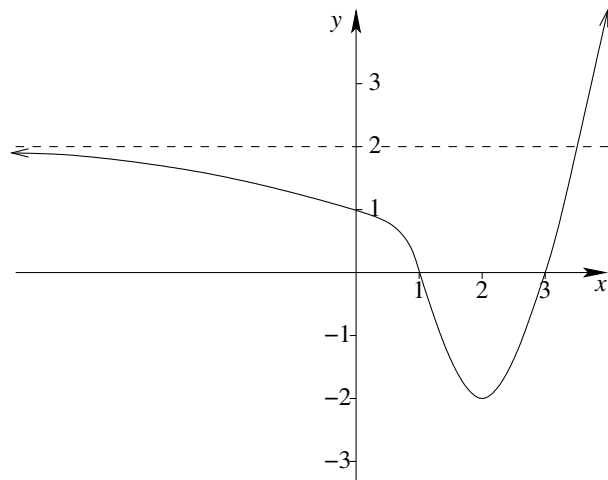
2



Marks

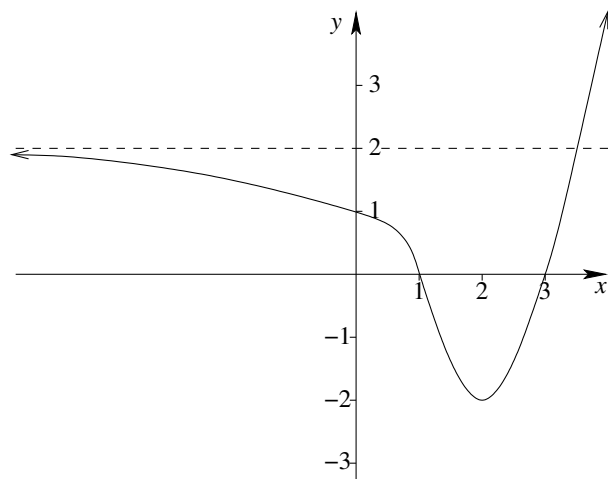
(ix) Sketch $|y| = f(x)$.

2



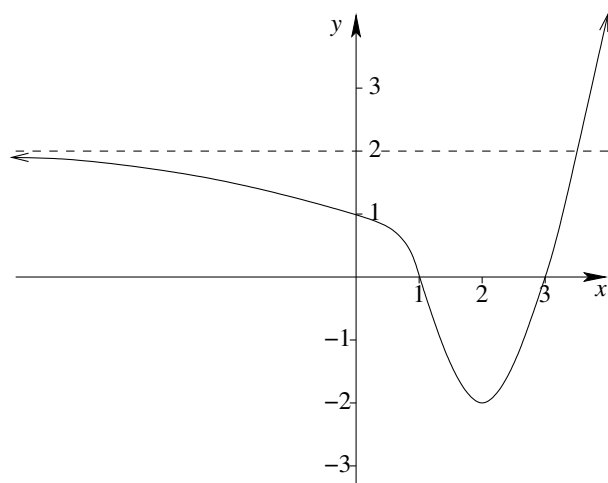
(x) Sketch $y = \frac{1}{f(x)}$.

2



(xi) Sketch $y = \ln f(x)$.

2





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**HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK # 2**

Mathematics Extension 2

Sample Solutions

2002 Ext 2 Task #2

$$(1) (a) \int \frac{1}{x \ln x} dx = \int \frac{\frac{1}{x}}{\ln x} dx \quad (\text{let } u = \ln x)$$

$$= \ln |\ln x| + C$$

$$(b) \int_0^{\pi/3} \sin^3 x \cos x dx$$

$$\begin{cases} \text{let } u = \sin x \Rightarrow du = \cos x dx \\ x=0 \Rightarrow u=0 \\ x=\pi/3 \Rightarrow u = \frac{\sqrt{3}}{2} \end{cases}$$

$$= \int_0^{\sqrt{3}/2} u^3 du$$

$$= \left[\frac{1}{4} u^4 \right]_0^{\sqrt{3}/2} = \frac{1}{4} \left[\left(\frac{\sqrt{3}}{2} \right)^4 - 0 \right]$$

$$= \frac{1}{4} \times \frac{9}{16} = \frac{9}{64}$$

Table of Integrals

$$(c) \int \frac{dx}{\sqrt{x^2+4x+8}} = \int \frac{dx}{\sqrt{(x+2)^2+4}} \quad \left[\int \frac{dx}{\sqrt{x^2+a^2}} = \ln(x + \sqrt{x^2+a^2}) \right]$$

$$= \ln |x+2 + \sqrt{x^2+4x+8}| + C$$

$$(d) \int_0^{\frac{1}{2}} \cos^{-1} x dx = \int_0^{\frac{1}{2}} 1 \times \cos^{-1} x dx$$

$$= \left[x \cos^{-1} x \right]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} x \times \frac{-1}{\sqrt{1-x^2}} dx$$

$$= \frac{1}{2} \cos^{-1} \left(\frac{1}{2} \right) - \int_0^{\frac{1}{2}} \frac{-x}{\sqrt{1-x^2}} dx = \frac{1}{2} \times \frac{\pi}{3} - \frac{1}{2} \int_0^{\frac{1}{2}} \frac{-2x}{\sqrt{1-x^2}} dx$$

$$= \frac{\pi}{6} - \left[\frac{1}{2} \times 2 \sqrt{1-x^2} \right]_0^{\frac{1}{2}}$$

$$= \frac{\pi}{6} - \left(\sqrt{1-\frac{1}{4}} - \sqrt{1} \right)$$

$$= \frac{\pi}{6} + 1 - \frac{\sqrt{3}}{2}$$

[OR let $u = 1-x^2$]

$$(e) \text{ (i) } \frac{1}{(x+2)(2x+1)} = \frac{A}{(x+2)} + \frac{B}{(2x+1)}$$

$$\therefore 1 \equiv A(2x+1) + B(x+2)$$

$$x = -\frac{1}{2} \Rightarrow 1 = 0 + B \times \frac{3}{2}$$

$$\therefore B = \frac{2}{3}$$

$$\because A + 2B = 1 \Rightarrow A + \frac{4}{3} = 1$$

$$\therefore A = -\frac{1}{3}$$

$$\begin{aligned} \int_0^1 \frac{dx}{(x+2)(2x+1)} &= \int_0^1 \left(\frac{-\frac{1}{3}}{x+2} + \frac{\frac{2}{3}}{2x+1} \right) dx \\ &= -\frac{1}{3} \int_0^1 \left(\frac{1}{x+2} - \frac{2}{2x+1} \right) dx = -\frac{1}{3} \left[\ln\left(\frac{x+2}{2x+1}\right) \right]_0^1 \end{aligned}$$

$$= -\frac{1}{3} \left[\ln\left(\frac{3}{3}\right) - \ln 2 \right]$$

$$= \frac{1}{3} \ln 2$$

$$(ii) \int_0^{\pi/2} \frac{3}{4+5\sin x} dx$$

$$= \int_0^1 \frac{3}{4+5\left(\frac{2u}{1+u^2}\right)} \times \frac{2du}{1+u^2}$$

$$= \int_0^1 \frac{3(1+u^2)}{(4(1+u^2)+10u)(1+u^2)} \times 2du$$

$$= \int_0^1 \frac{3du}{(2u^2+5u+2)}$$

$$= 3 \int_0^1 \frac{du}{2u^2+5u+2} = 3 \int_0^1 \frac{du}{(u+2)(2u+1)}$$

$$= 3 \times \frac{1}{3} \ln 2 \quad [\text{from (i)}]$$

$$= \ln 2$$

$$\text{let } u = \tan x/2$$

$$\therefore du = \frac{1}{2} \sec^2 x/2 dx$$

$$\therefore dx = \frac{2}{\sec^2 x/2} du$$

$$= \frac{2}{1+u^2} du$$

$$\sin x = \frac{2u}{1+u^2}$$

$$x=0 \Rightarrow u=0$$

$$x=\pi/2 \Rightarrow u=1$$

$$(f) f(x) = x^8 \sin x$$

$$\therefore f(-x) = (-x)^8 \sin(-x)$$

$$= x^8 \times -\sin x$$

$$= -x^8 \sin x$$

$$= -f(x) \therefore \text{ODD}$$

$$\therefore \int_{-\pi/2}^{\pi/2} x^8 \sin x dx = 0$$

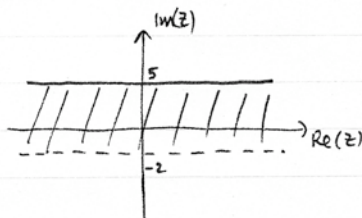
$$\begin{aligned} (2) \quad (a) \quad (i) \quad i^{2002} &= (i^2)^{1001} \\ &= (-1)^{1001} \\ &= -1 \end{aligned}$$

$$(ii) \quad 2z^2 + (3+i)z + 2 = 0$$

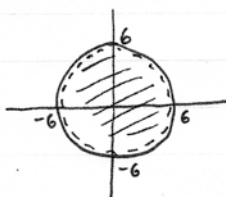
$$\begin{aligned} \therefore z &= \frac{-(3+i) \pm \sqrt{(3+i)^2 - 4 \times 2 \times 2}}{4} \\ &= \frac{-(3+i) \pm \sqrt{8+6i-16}}{4} \\ &= \frac{-3-i \pm \sqrt{6i-8}}{4} \\ &= \frac{-3-i \pm (1+3i)}{4} \\ &= \frac{-2+2i}{4}, \frac{-4-4i}{4} \\ &= \frac{-1+i}{2}, -(1+i) \end{aligned}$$

$$\begin{aligned} (x+iy)^2 &= -8+6i \\ \therefore x^2-y^2 &= -8 \\ 2xy &= 6 \\ 1-y &= -8 \\ \therefore \boxed{x=1, y=3} \end{aligned}$$

$$(b) \quad (i) \quad -2 < \operatorname{Im}(z) \leq 5$$



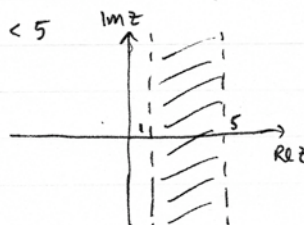
$$(ii) \quad |z| < 6$$



$$(iii) \quad 2 < z + \bar{z} < 10$$

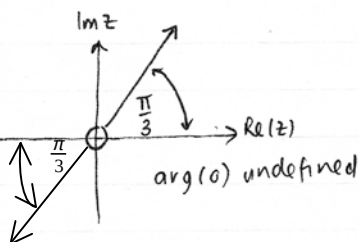
$$\therefore 2 < 2\operatorname{Re} z < 10$$

$$\therefore 1 < \operatorname{Re} z < 5$$



$$\begin{aligned} (iv) \quad \arg(z^2) &= \frac{2\pi}{3} \\ \therefore 2\arg z &= \frac{2\pi}{3} \\ \therefore \arg z &= \frac{\pi}{3} \end{aligned}$$

$$\arg z = \frac{\pi}{3}, -\frac{2\pi}{3}$$



The two solutions
because of z^2

(2) (c) $\binom{10}{3}$ or $\binom{10}{7}$

(d) $x^3 - 2x^2 + 2x - 2 = 0$

$\alpha + \beta + \gamma = 2$

$\alpha\beta + \alpha\gamma + \beta\gamma = 2$

$$\begin{aligned} \therefore \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= 4 - 2(2) \\ &= 0 \end{aligned}$$

sum of squares is zero, so non-real roots involved
BUT since the coefficients are real, the non-real roots occur in conjugate pairs. So only one real root

e) $P(x)$ is odd polynomial deg 5

$\therefore P(x) = ax^5 + bx^3 + cx$

$P(1) = P(2) = 0$ and $P(0) = 0$

$\therefore P(x) = x(x-1)(x-2)Q(x)$, $\deg Q(x) = 2$

BUT $P(x)$ is odd $P(1) = 0 \Rightarrow P(-1) = 0$
 $P(2) = 0 \Rightarrow P(-2) = 0$

$\therefore P(x) = ax(x+1)(x-1)(x+2)(x-2)$

$P(3) = 240$

$\Rightarrow 3a(4)(2)(5)(1) = 240$

$\Rightarrow 120a = 240$

$\Rightarrow a = 2$

$\therefore P(x) = 2x(x+1)(x-1)(x+2)(x-2)$

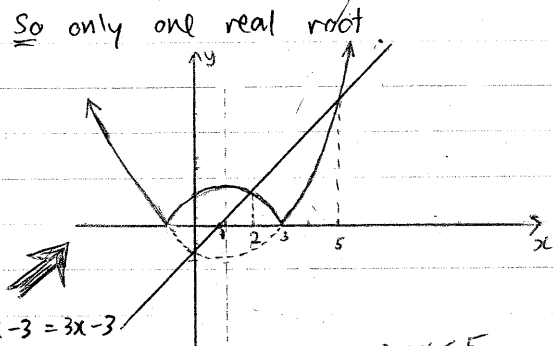
(3) (a) $|x^2 - 2x - 3| < 3x - 3$

or $|(x-3)(x+1)| < 3(x-1)$

Find points of intersection

check $-(x^2 - 2x - 3) = 3x - 3$
 $-x^2 - x + 6 = 0$
 $x^2 + x - 6 = 0$
 $(x+3)(x-2) = 0$

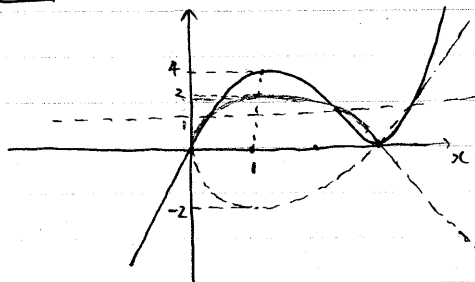
$x^2 - 2x - 3 = 3x - 3$
 $\therefore x^2 - 5x = 0$
 $x(x-5) = 0$
 $x = 5$



$\therefore 2 < x < 5$

(b) $y = x(x-3)^2$
 $= x(x^2 - 6x + 9)$
 $= x^3 - 6x^2 + 9x$
 $y' = 3x^2 - 12x + 9$
 $= 3(x^2 - 4x + 3)$
 $= 3(x-1)(x-3)$

$x = 1, y = 4$



The diagrams on this sheet each show a graph of the function $y = f(x)$, as shown on page 4 of your question booklet. Using the given graphs as a guide, sketch the required graphs.

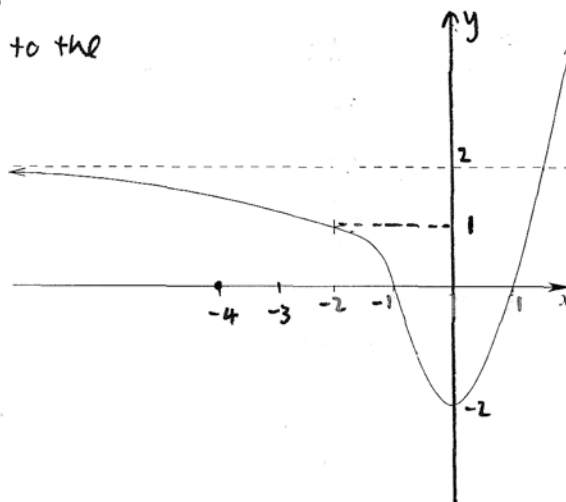
Insert this sheet into your answer booklet for Question 3.

Mark

(vii) Sketch $y = f(x+2)$,

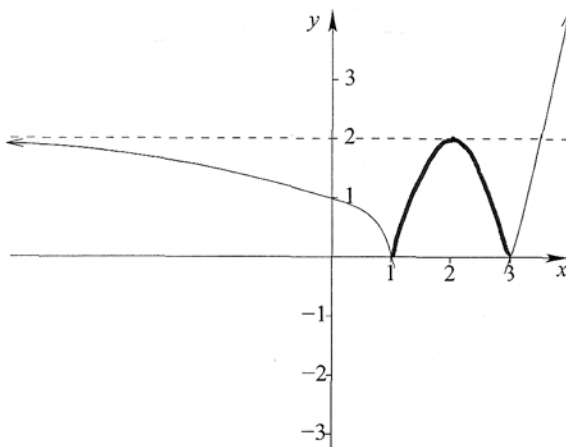
2

move y-axis 2 units to the right



(viii) Sketch $y = |f(x)|$.

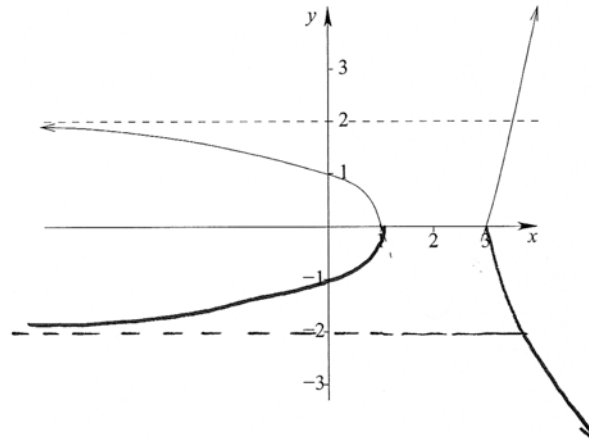
2



Marks

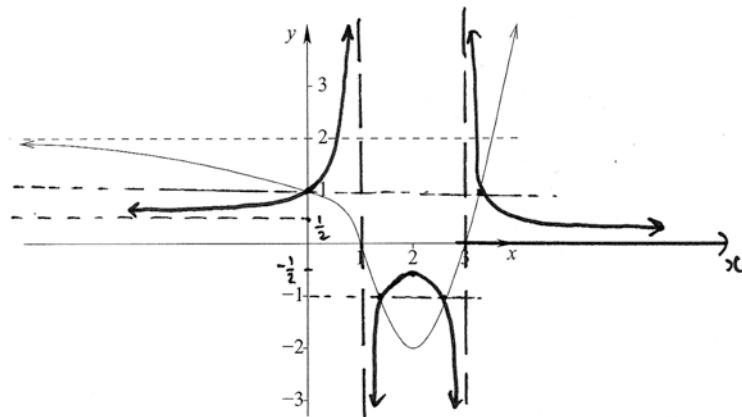
(ix) Sketch $|y| = f(x)$.

2



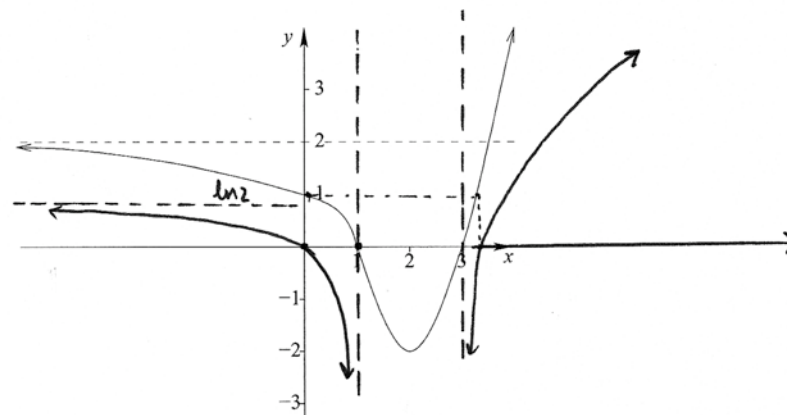
(x) Sketch $y = \frac{1}{f(x)}$.

2



(xi) Sketch $y = \ln f(x)$.

2



(4)

$$(a) \int_0^{\frac{1}{2}} \frac{\sqrt{x}}{(1-x^2)^{3/2}} dx$$

$$= \int_0^{\pi/4} \frac{\sin \theta \cdot 2 \sin \theta \cos \theta d\theta}{(1 - \sin^2 \theta)^{3/2}}$$

$$= \int_0^{\pi/4} \frac{2 \sin^2 \theta \cos \theta d\theta}{\cos^3 \theta}$$

$$= 2 \int_0^{\pi/4} \tan^2 \theta d\theta$$

$$= 2 \int_0^{\pi/4} (\sec^2 \theta - 1) d\theta$$

$$= 2 \left[\tan \theta - \theta \right]_0^{\pi/4}$$

$$= 2 \left[1 - \pi/4 \right]$$

$$= 2 - \pi/2$$

$$x = \sin^2 \theta$$

$$\therefore dx = 2 \sin \theta \cos \theta d\theta$$

$$x=0 \Rightarrow \theta=0$$

$$x=\frac{1}{2} \Rightarrow \theta=\pi/4$$

$$* \sqrt{x} = \sqrt{\sin^2 \theta} = |\sin \theta|$$

$$= \sin \theta \text{ for } 0 \leq \theta \leq \frac{\pi}{4}$$

[N.B. if you chose $\theta = -\pi/4$ above then you need to choose $-\sin \theta$]

$$\sqrt{\cos^2 \theta} = \cos \theta \text{ for } 0 \leq \theta \leq \pi/4$$

$$(b) f(x) = f(a-x)$$

$$(i) \int_0^a x f(x) dx$$

$$\text{let } u = a-x \Rightarrow du = -dx$$

$$x=0 \Rightarrow u=a$$

$$x=a \Rightarrow u=0$$

$$x=a-u$$

$$= \int_a^0 (a-u) f(a-u) (-du)$$

$$= \int_0^a (a-u) f(a-u) du$$

$$\therefore \int_0^a x f(x) dx = \int_0^a a f(a-x) dx - \int_0^a x f(a-x) dx$$

$$= \int_0^a a f(x) dx - \int_0^a x f(x) dx$$

$$\therefore 2 \int_0^a x f(x) dx = a \int_0^a f(x) dx$$

$$\therefore \int_0^a x f(x) dx = \frac{a}{2} \int_0^a f(x) dx$$

$$(ii) \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$$

$$[u = \cos x]$$

$$= \frac{\pi}{2} \int_1^{-1} \frac{-du}{1+u^2} = 2 \times \frac{\pi}{2} \times \int_0^1 \frac{du}{1+u^2} \quad (\text{even})$$

$$= \pi \times \tan^{-1}(1) = \pi \times \pi/4 = \pi^2/4 \quad \text{Q.E.D.}$$

-8-

$$\begin{aligned} (4) (c) (i) \quad I_n &= \int_0^1 x^n e^{2x} dx \\ &= \left[\frac{1}{2} e^{2x} \times x^n \right]_0^1 - \int_0^1 \left(\frac{1}{2} e^{2x} \right) \times n x^{n-1} dx \\ &= \left(\frac{1}{2} e^2 \times 1 \right) - (0) - \frac{n}{2} \int_0^1 x^{n-1} e^{2x} dx \\ &= \frac{e^2}{2} - \frac{n}{2} I_{n-1} \end{aligned}$$

$$I_n = \frac{1}{2} (e^2 - n I_{n-1})$$

$$(ii) \quad I_4 = \int x^4 e^{2x} dx$$

$$I_4 = \frac{1}{2} (e^2 - 4 I_3)$$

$$I_3 = \frac{1}{2} (e^2 - 3 I_2)$$

$$I_2 = \frac{1}{2} (e^2 - 2 I_1)$$

$$I_1 = \frac{1}{2} (e^2 - I_0)$$

$$I_0 = \int_0^1 e^{2x} dx = \frac{1}{2} e^{2x} \Big|_0^1 = \frac{1}{2} (e^2 - 1)$$

$$I_1 = \frac{1}{2} e^2 - \frac{1}{2} (e^2 - 1) = \frac{1}{4} (e^2 + 1)$$

$$I_2 = \frac{1}{2} e^2 - 2 \times \frac{1}{4} (e^2 + 1) = \frac{1}{4} (e^2 - 1)$$

$$I_3 = \frac{1}{2} e^2 - 3 \times \frac{1}{4} (e^2 - 1) = \frac{1}{8} (e^2 + 3)$$

$$I_4 = \frac{1}{2} e^2 - 4 \times \frac{1}{8} (e^2 + 3) = \frac{1}{4} (e^2 - 3)$$

(5)

$$(i) \quad y = x^2 - b \quad y = \frac{k}{x}$$

$$\therefore x^2 - b = \frac{k}{x}$$

$$\therefore x^3 - bx = k$$

$$\therefore x^3 - bx - k = 0$$

(ii) A is where they touch i.e. a common tangent.
Hence the double root, since there must be 3 solutions, so at A they are identical.

$$(iii) \quad \text{let } f(x) = x^3 - bx - k$$

$$\therefore f'(x) = 3x^2 - b$$

let $x = \alpha$ be the x -coord of A

$$\therefore f(\alpha) = f'(\alpha) = 0$$

$$\therefore 3\alpha^2 = b \quad (b > 0)$$

$$f(\alpha) = 0 \Rightarrow \alpha(\alpha^2 - b) = k$$

$$\therefore \alpha^2(\alpha^2 - b) = k^2$$

$$\therefore \frac{b}{3} \left(\frac{b}{3} - b \right)^2 = k^2$$

$$\therefore \frac{b}{3} \times \left(-\frac{2b}{3} \right)^2 = k^2$$

$$\therefore \frac{b}{3} \times \frac{4b^2}{9} = k^2$$

$$\therefore 4b^3 = 27k^2$$

$$(iv) \quad b = 12 \Rightarrow 4 \times 12^3 = 27k^2$$

$$\therefore 4 \times 1728 = 27k^2$$

$$\therefore k^2 = 256$$

$$k > 0 \Rightarrow k = 16$$

$$\therefore \alpha + \alpha + \beta = 0 \Rightarrow 2\alpha + \beta = 0 \quad (1) \quad (\beta \text{ is the } x\text{-coord of B})$$

$$\alpha^2 + 2\alpha\beta = -b \Rightarrow \alpha^2 + 2\alpha\beta = -12$$

$$\alpha^2\beta = -(-k) \Rightarrow \alpha^2\beta = 16 \quad (2)$$

$$(1) \Rightarrow (2) \quad \beta = -2\alpha \Rightarrow \alpha^2(-2\alpha) = 16$$

$$-\alpha^3 = 8 \Rightarrow \alpha = -2 \quad \therefore \beta = 4$$

$$\therefore A(-2, -8) \quad B(4, 4)$$

5(b) 10 R, 10 B, 10 Y

$R_1, \dots, R_{10}, B_1, \dots, B_{10}, Y_1, \dots, Y_{10}$

(i) 4 cards $\Rightarrow \binom{30}{4} = 27405$

No red card $\Rightarrow \binom{20}{4} = 4845$

preferred
solution

$\therefore$ At least one red card $= {}^{30}C_4 - {}^{20}C_4 = 22560$

OR $\binom{10}{1}\binom{20}{3} + \binom{10}{2}\binom{20}{2} + \binom{10}{3}\binom{20}{1} + \binom{10}{4}$

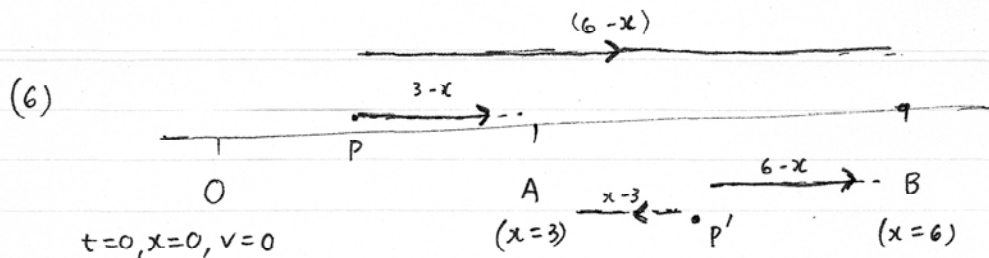
(ii) At least one Red, and at least one of each colour

$\therefore$ RRYB, RYYB, RYBB

e. $\binom{10}{2}\binom{10}{1}\binom{10}{1} \times 3$

$= 13500$

$\therefore$ Prob $= \frac{13500}{22560} = \frac{225}{376} \doteq 59.8\%$



(arrows indicating direction of acceleration)

(i) For $0 \leq x \leq 3$ or for some point P :

$$\begin{aligned}
 \ddot{x} &= (3-x) + (6-x)^2 \\
 &= 3-x + 36-12x+x^2 \\
 &= x^2-13x+39
 \end{aligned}$$

For $3 \leq x \leq 6$ or for some point P'

N.B. distance from A is $x-3$ BUT the acceleration is negative
distance from B is $6-x$ BUT acceleration is positive.

$$\begin{aligned}
 \therefore \ddot{x} &= -(x-3) + (6-x)^2 \\
 &= x^2-13x+39
 \end{aligned}$$

$$(ii) \quad \ddot{x} = \frac{d(\frac{1}{2}v^2)}{dx} = x^2-13x+39$$

$$\therefore \frac{1}{2}v^2 = \frac{1}{3}x^3 - \frac{13}{2}x^2 + 39x + C$$

$$\begin{aligned}
 v^2 &= \frac{2}{3}x^3 - 13x^2 + 78x + k \quad (x=0, v=0) \\
 &\Rightarrow k=0
 \end{aligned}$$

$$\begin{aligned}
 \therefore v^2 &= \frac{2}{3}x^3 - 13x^2 + 78x \\
 &= \frac{2}{3}(2x^2 - 39x + 234)
 \end{aligned}$$

$$(iii) \quad v=0 \Rightarrow \frac{x}{3}=0 \text{ or } 2x^2-39x+234=0$$

$$\therefore x=0 \quad \text{or} \quad 2x^2-39x+234=0$$

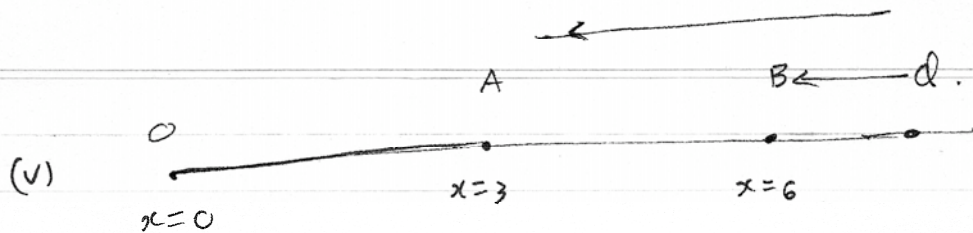
$$\text{BUT } \Delta = -351 < 0$$

$\therefore$ no real solution

or $v \neq 0$ except at $x=0$ re. initially

$$\begin{aligned}
 (iv) \quad x=6, \quad \therefore v^2 &= \frac{2}{3}(2 \times 36 - 39 \times 6 + 234) \\
 &= 2(72) = 144
 \end{aligned}$$

$$\therefore \text{speed} = |v| = 12$$



distance from A is $x-3$

distance from B is $x-6$

but acceleration is towards A and B

$$\begin{aligned}\therefore \ddot{x} &= -(x-3) - (x-6)^2 \\ &= -x+3 - (x^2-12x+36) \\ &= -x+3 - x^2+12x-36 \\ &= -x^2+11x-33\end{aligned}$$

$$(vi) \quad \therefore \frac{d(\frac{1}{2}v^2)}{dx} = -x^2+11x-33$$

$$\frac{1}{2}v^2 = -\frac{1}{3}x^3 + \frac{11}{2}x^2 - 33x + C$$

$$\therefore v^2 = -\frac{2}{3}x^3 + 11x^2 - 66x + K \quad (x=6, v^2=144)$$

$$144 = -144 + 11 \times 36 - 66 \times 6 + K$$

$$\therefore K = 288$$

N.B. $v^2 = -\frac{2}{3}x^3 + 11x^2 - 66x + 288$

$$\text{at } x=11: \quad v^2 = -\frac{2}{3} \times 11^3 + 11 \times 11^2 - 66 \times 11 + 288 = 5\frac{2}{3}$$

$\therefore$ particle is IN motion at $x=11$

$$\text{at } x=12: \quad v^2 = -\frac{2}{3} \times 12^3 + 11 \times 12^2 - 66 \times 12 + 288 = -72$$

$\therefore$ it does NOT reach $x=12$

So it MUST stop $11 < x < 12$ Q.E.D.